2022 exam - GLM

Thursday, 27 October 2022 4:04 pm

$X_1 .. \quad X_n \sim exp(rate = \frac{1}{\lambda})$

$$p(x|\lambda) = \frac{1}{\lambda} e^{-\frac{x}{\lambda}}$$

(a)

$$b_g(\lambda) = b_g\, p(x_1 .. x_n|\lambda)$$

$$= -n b_g \lambda - \frac{1}{\lambda} \Sigma x_i \quad (a)$$

$$\frac{\partial b_g(\lambda)}{\partial \lambda} = \frac{-n}{\lambda} + \frac{1}{\lambda^2} \Sigma x_i = 0$$

$$-n\lambda + \Sigma x_i = 0$$

$$\hat{\lambda} = \frac{\Sigma x_i}{n} \quad (c)$$

$$\frac{\partial^2 b_g(\lambda)}{\partial \lambda^2} = \frac{n}{\lambda^2} - \frac{2}{\lambda^3} \Sigma x_i$$

$$I(\lambda) = -E\left(\frac{n}{\lambda^2} - \frac{2}{\lambda^3} \Sigma x_i\right)$$

$$= \frac{-n}{\lambda^2} + \frac{2}{\lambda^3} n\lambda$$

$$= \frac{-n + 2n}{\lambda^2} = \frac{n}{\lambda^2} \quad (b)$$

$$\hat{\lambda} \sim N\left(\lambda, \frac{\lambda^2}{n}\right) \sim (c)$$